

Assessment of the tomato cluster yield estimation algorithms via tracking-by-detection approaches

Zhongxian Qi^a, Tianxue Zhang^{b,c,*}, Ting Yuan^{a,*}, Wei Zhou^a, Wenqiang Zhang^{a,*}

^a College of Engineering, China Agricultural University, Beijing 100083, China

^b School of Mechanical Engineering and Automation, Beihang University, Beijing 100191, China

^c Institute of Medical Robotics, Shanghai Jiaotong University, Shanghai 200240, China

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ABSTRACT

Automated vision-based detection and counting are critical for accurate tomato yield estimation, which contribute to precise yield management strategies and an efficient food supply chains. Special conditions, including background clutter, occlusion, and varying sunlight, affect the accuracy of crop detection and counting. To determine the most suitable algorithms for this yield estimation context, we herein establish a public multi-object tracking (MOT) dataset for tomato cluster counts, while evaluating and comparing state-of-the-art target detection and MOT-based algorithms. The evaluated detectors consist of YOLOv8 and RT-DETR, which represent algorithms that achieve a balance between accuracy and speed. The tracking algorithms included state-of-the-art methodologies such as SORT, DeepSort, ByteTrack, and BotSort. Initially, the performance of the detectors was rigorously evaluated, followed by a comprehensive assessment of the four tracking algorithms within a multi-target tracking database tailored for this research and structured in the MOT context. The findings reveal that YOLOv8 and RT-DETR achieve 93.6% and 94.9% results at mAP@75, respectively, with RT-DETR exhibiting fewer false detections. When combined with the RT-DETR detector, the ByteTrack-based algorithm registers the highest counting accuracy at 95.5%, whereas BotSort achieves the highest MOTA score with 84.6%. Notably, the trackers without the ReID module (i.e., SORT and ByteTrack) demonstrate greater adaptability to frame rate variations in the test videos. At a 30-fps frame rate, the incorporation of ReID modules in DeepSort and BotSort algorithms significantly enhances the MOTA metric. Looking ahead, we plan to leverage these algorithms into an autonomous inspection platform that aims to estimate crop yield in real-time.

1. Introduction

With the development of modern facility agriculture, the greenhouse fruit and vegetable planting pattern has attracted much attention due to its stable and reliable yield and quality [20,21,24]. This planting pattern allows for close collaboration with the food supply chain, that is, tailoring the yield according to market demand. Significantly, accurate yield estimation constitutes a pivotal linkage in building an efficacious food supply chain. Tomatoes, a ubiquitously cultivated fruit and vegetable, have gradually increased the level of greenhouse cultivation and area in recent years [26]. The manual counting of tomato quantities is laborious and time-consuming. In contrast, the use of computer vision technology to estimate tomato yield as an automated and efficient approach to reduce labor costs is currently the predominant methodology for determining tomato population statistics [1,23].

Object detection technology is one of the most widely used vision technologies [16,34,37]. Research in this field has evolved from manual feature design to the use of deep learning-based enhancements that show superior performance. This approach is capable of precisely separating elements such as fruits and crops from the background and accomplishing the localization of each object, providing a reasonable basis for counting fruits [2,6]. Constantly evolving detectors have helped in the accurate detection of fruits and crops in unstructured agricultural environments. Deep learning-based detectors have evolved from a two-stage process to a single-stage one. Single-stage detectors can be distinguished by the underlying deep learning architecture, such as those constructed on the basis of convolutional neural networks (CNN) and transformer architectures. It is noteworthy that single-stage detectors based on the CNN architecture, as demonstrated by the YOLO series, have achieved trade-off between accuracy and efficiency

* Corresponding authors.

E-mail addresses: tianxuezhang@buaa.edu.cn (T. Zhang), yuanting122@cau.edu.cn (T. Yuan), zhangwq@cau.edu.cn (W. Zhang).

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[8,9,22,29]. In contrast, detectors utilizing transformer architecture offer superior accuracy; however, their application in real-time detection is constrained by significantly computational costs [4]. Recent some studies have overcome the performance shortcomings of transformer architecture [15,18]. It is also of importance mentioning that due to the diversity of the growing environment and agronomic conditions of crops, object detection methods often should to be combined with other techniques. For instance, Zhang, et al. [36] employed the SIFT image-matching algorithm and the YOLOX detector to achieve detection and counting with robustness against shadows, occlusions, and incomplete outlines. Xiong et al. [32] adopted drone technology in conjunction with object detection algorithms to achieve accurate detection of the surface of young mangoes in tree canopies. The primary objective of current study is to conduct a thorough assessment of counting and estimation of tomato yield under ridge planting conditions. This is effectively performed by acquiring a sequence of images (or videos) from each planting row, which presents complex technical challenges in cross-frame object detection and identification.

Several studies have applied video tracking methodologies to estimate crop yield in agricultural scenarios and have achieved good results [2,7,19,25,31,33]. In the field of video tracking technology, multi-object tracking (MOT) is capable effectively dealing with the identification of targets that appear multiple times in video frames [11,17]. Among them, the tracking-by-detection strategy is an effective approach for MOT. The main part is to first locate the tomato clusters in the video frames through a target detector, and then use a tracker to assign the same identification number to the same targets in different frames. These identification numbers can be effectively utilized as the basis for fruit cluster counting. In such a framework, both the object detector and the tracker are key factors affecting the counting accuracy (X. [13,27]. The accuracy of the detector directly affects the performance of the tracker, which focuses on the data association of the target objects in the frames. The corresponding trackers based on the Kalman filter algorithm and the Hungarian matching algorithm are a class of promising algorithms suitable for crops and fruit counting. Some studies have compared the potential applications of these algorithms with other algorithms and highlighted their significant advantages [1,28,31,33]. However, some former investigations have mainly focused on improving the detector output results (P. [12,23,31]. The existing literature lacks an analysis of the challenges and limitations of applying various tracking algorithms in agricultural scenarios, and there is also a lack of a systematic evaluation of the performance and robustness of these algorithms. By this view, study also aims to compare the performance discrepancies of several advanced trackers using two representative state-of-the-art target detectors for counting and estimating tomato yield. Simultaneously, we focus on the challenges fronting video-based tomato cluster counting, which two major ones of them are as:

- *Interference from the background row of tomato clusters.* Detectors are able to detect as many fruit clusters as possible in the image but this approach does not achieve the goal of accurate counting. In estimating yield performance, it is necessary to accurately count the number of fruit clusters in the current row without considering those in the background rows. Therefore, we only mark and train the fruit clusters located in the current row in the image, and this presents a significant challenge for the detector.
- *Occlusion, appearance similarity, and sunlight interference.* Some trackers measure whether the detected fruit clusters in the current frame and the previous frames are the same tomato by comparing their appearance. However, occlusion, similar appearance, and lighting discrepancies under various angles could cause certain interferences in comparing of the appearance of tomatoes.

In addition to the above two main issues, there is also no public database to analyze the impact of advanced detectors on trackers and compare the performance of MOT-based algorithms in fruit bunch

counting. On these bases, we originally contribute to the following issues:

- Creating a public dataset for MOT of tomato clusters.¹
- Evaluating the performance discrepancies of advanced detectors and trackers in agricultural environments.
- Examining the prospects of various MOT algorithms in the application of tomato cluster counting.

The organization of this research work is mainly as follows: Section 2 introduces the utilized databases, advanced detectors, and tracking algorithms; Section 3 presents the results and discussion obtained from the sensitivity analysis and application tasks; and finally, Section 4 presents the main results obtained from this investigation.

2. Materials and methods

This study develops and compares the performance of advanced object detectors and multi-target trackers in a tomato cluster yield estimation system and presents its overall structure as illustrated in Fig. 1. The data were captured via a camera fixed on mobile platforms or unmanned vehicles. Subsequently, the data were divided into two main categories: images and videos, which were utilized to evaluate the performance of detectors and trackers, respectively. The process presented in the dashed black box has effectively evaluated the performance of the target detectors. On this basis, the working principle of the detectors can be summarized in three main steps: feature extraction, feature encoding, and feature decoding, with different detectors having distinct implementations in these stages. Additionally, the procedure marked with the green box in the figure is adopted to assess and compare the performance of various trackers in tomato counting. The working principle of the trackers is illustrated in Fig. 4, and a detailed description has been provided in Section 2.3.

2.1. Data preparation

2.1.1. Data acquisition

The experimental datasets (i.e., original images and original videos) were collected in Hongfu Agricultural Tomato Production Park, Daxing District, Beijing (39.603538°N 116.283601°E). The greenhouse adopts a high-yield, and high-input Dutch greenhouse planting pattern. The agronomic conditions are row-row planting with a spacing of 1600 mm between every two rows and approximately 10–12 kernels per tomato bunch. To maintain a fixed growth height of the tomato plants, the vines are artificially dropped at regular intervals. This pattern allows the tomato clusters in the color-turning and mature stages to be distributed on the main stems with a vertical height range of 1400 mm (see Fig. 2a,b). Some researchers have utilized Realsense or Azure Kinect cameras to collect data; however, the relatively small field of view of these cameras makes it difficult to fully cover the tomatoes at the color-turning and mature stages in narrow crop rows. Therefore, we employed the OAK-D-Pro-W camera [14] with a wide field of view (FOV) as a data collection device. The camera's horizontal and vertical FOV of the camera are 95° and 70°, respectively. After testing, the images obtained by the camera could completely cover the tomatoes in at the coloring and ripening stages in this scene, as shown in Fig. 2c. The camera was mounted on a platform vehicle moving along the track. The platform vehicle moves along the track at a speed of about 0.3 m/s. We collected 15 raw videos from 2023 to 2024 to test the performance of the tracker. For the detector, we captured several sequential videos and extracted them at one every 15 frames to form a dataset for target detection. In this way, 1700

¹ The Tomato cluster MOT dataset utilized here is publicly available through the following link: <https://pan.baidu.com/s/1FJMs5jgJFjSGHBpu7zu29g?pwd=37yu>.

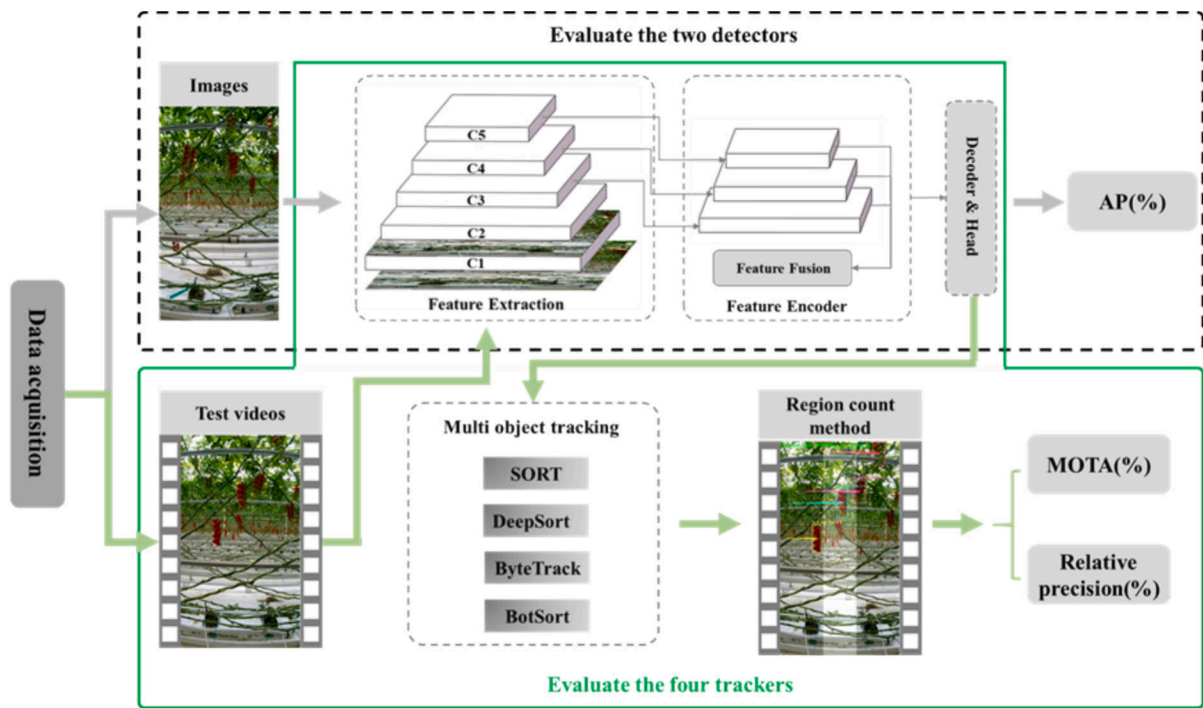


Fig. 1. General flowchart of greenhouse tomato clusters counting via the tracking by detection strategy. First, the data is acquired and categorized into image data and video data. Image data and video data are then effectively utilized to assess the performances of the detectors and trackers, respectively. The process mentioned in the black dashed box in the figure is also implemented to compare the performance of the two detectors. Also, the process presented within the green-boxed line is utilized to evaluate the performance of the trackers.

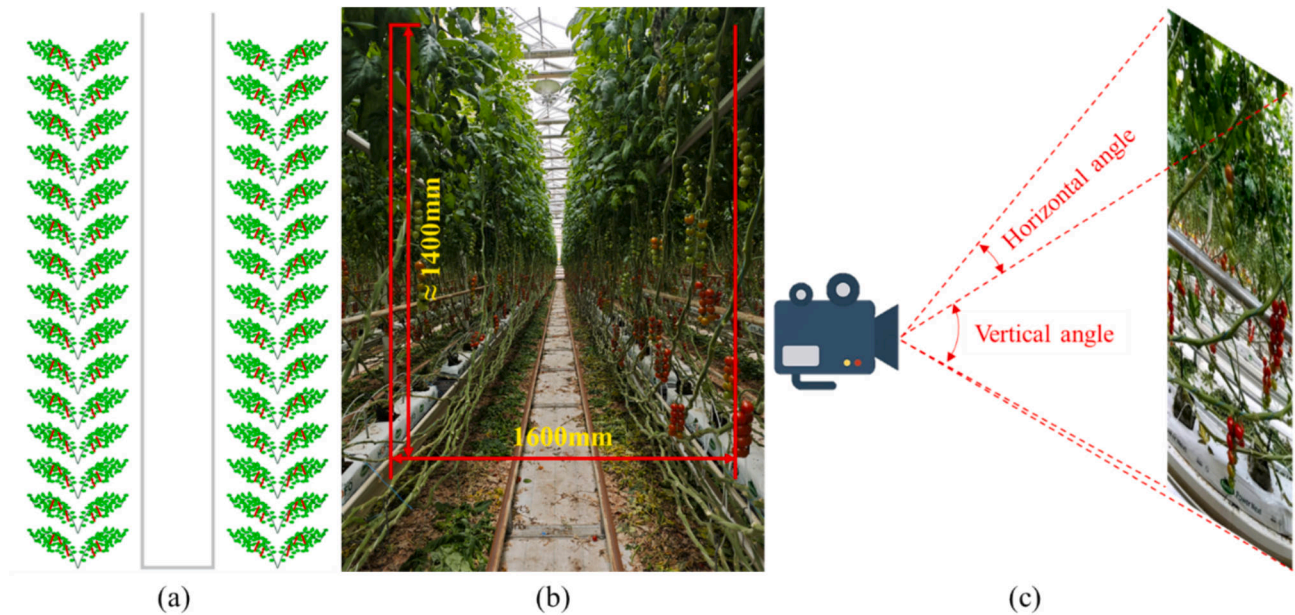


Fig. 2. Schematic representation of the growing area of the tomato bunches and the camera's field of view: (a) Top view of row for tomato planting, (b) Tomato growing zone, (c) A general scope of the camera field of view.

images were collected. Since some tracking algorithms such as DeepSort require separate training for the feature extractor, we utilized the sequence bounding box extracted from the videos used for the target detection dataset to train the feature extractor. The resolution of the images and videos is 720 * 1280 and the video frame rate is 30 fps.

2.1.2. Dataset building

The dataset utilized for the bunches detection models is made up of

raw images that can be used for training and testing the object detection models. A total of 1700 raw images were randomly divided into a training dataset and a test dataset (8:2 division). A total of 9068 object instances were labeled. The dataset used for video tracking was semi-automatically labeled using an appropriate labeling software, namely DarkLabel [5]. The composition of the labeled video dataset has been presented in Table 1.

Table 1
Summary of databases utilized for counting tomato clusters.

Video name	Resolution	Frames	Object instances	Tomato clusters
VT1	720 × 1280	3227	10,220	112
VT2	720 × 1280	3120	11,236	117
VT3	720 × 1280	2565	7544	79
VT4	720 × 1280	3519	11,725	120
VT5	720 × 1280	2640	13,273	91
VT6	720 × 1280	2734	13,698	91
VT7	720 × 1280	2173	7435	86
VT8	720 × 1280	3012	10,129	119
VT9	720 × 1280	2301	8086	100
VT10	720 × 1280	2926	10,536	128
VT11	720 × 1280	1230	4943	52
VT12	720 × 1280	3572	11,747	140
VT13	720 × 1280	2262	13,875	115
VT14	720 × 1280	2498	14,543	123
VT15	720 × 1280	2747	16,091	141

2.2. Cluster detection

After constructing the dataset used for object detection, we briefly describe the detector architectures used in this work for tomato bunch detection. The performance of deep learning-based approaches on detection tasks outperformed classical computer vision algorithms. Fig. 3 demonstrates a schematic of the main architectures of a CNN-based and transformer-based single-stage target detectors. In current study, we aim to compare the performance discrepancies between the two types of detectors used for tomato detection and their impact on the MOT-based algorithm for future practical applications.

2.2.1. YOLOv8 based on the CNN architecture

The YOLO algorithm family of algorithms is a representative single-stage object detection algorithm based on the CNN architecture [22]. Due to the excellent trade-off between accuracy and efficiency, it has been widely used in the industry since its introduction. After many iterations by the community, it is now broadly employed in various agricultural scenarios [12,27,36]. In the present work, YOLOv8 is utilized as a representative of a single-stage detector to compare its performance with other detection algorithms [10]. In the YOLOv8 architecture, the features are first extracted by the backbone structured as a CNN model, then combined by the neck layer for different scales, and finally the results are output by the detection classification detector and regression detector headers, and then processed by the NMS to

obtain the final results. Additionally, YOLOv8 further optimizes the loss function, enabling the model to converge quickly and achieve better performance metrics.

2.2.2. RT-DETR based on the transformer architecture

In recent years, the accuracy of the transformer architecture detectors has exceeded that of CNN architecture; nevertheless, their memory consumption and slow operation remain a substantial drawback. Therefore, many efforts have been devoted to optimizing the transformer architecture, with RT-DETR [15] being a reprehensive example. RT-DETR first attempts to analyze various factors affecting the model's inference speed, such as post-processing NMS modules. It achieves efficient inference by mixing CNN and transformer architectures for processing. Additionally, the post-processing of the NMS is eliminated to achieve end-to-end detection results, which further reduces time loss. Its backbone also adopts the CNN model and employs an efficient hybrid coding approach to encode features from different layers. The decoding prediction head utilizes an IoU-aware query selection technique, which can directly output results without the need for NMS, simply by filtering through a specific threshold.

2.3. Cluster tracking

The tracking-by-detection algorithm, which essentially relies on a well-performing detector, is currently the dominant scheme for implementing MOT. According to the discrepancies in the subsequent processing, the tracking algorithms can be classified into two types: CNN appearance-based feature extraction and non-appearance-based feature extraction. The main components of the tracking algorithms are illustrated in Fig. 4. First, the bounding boxes are localized by a detector. In continuing, the appearance features of the targets are commonly extracted by an appropriate CNN network. Additionally, omitting the step in some algorithms improves the tracking speed [3,35], and the dashed box indicates that these trackers do not require this step of feature extraction. The next step is similarity comparison and computation, and finally, the data correlation is utilized to identify the bounding boxes of the target between various frames. We employ four common tracking algorithms to perform the tomato bunches counting task. All of these tracking algorithms belong to the tracking-by-detection framework, which relies on advanced detectors. These trackers achieve advanced performance and widely used in agriculture, yet there is a lack of comparative analysis regarding their performance. To bridge this scientific gap, this study aims to compare the performance of these

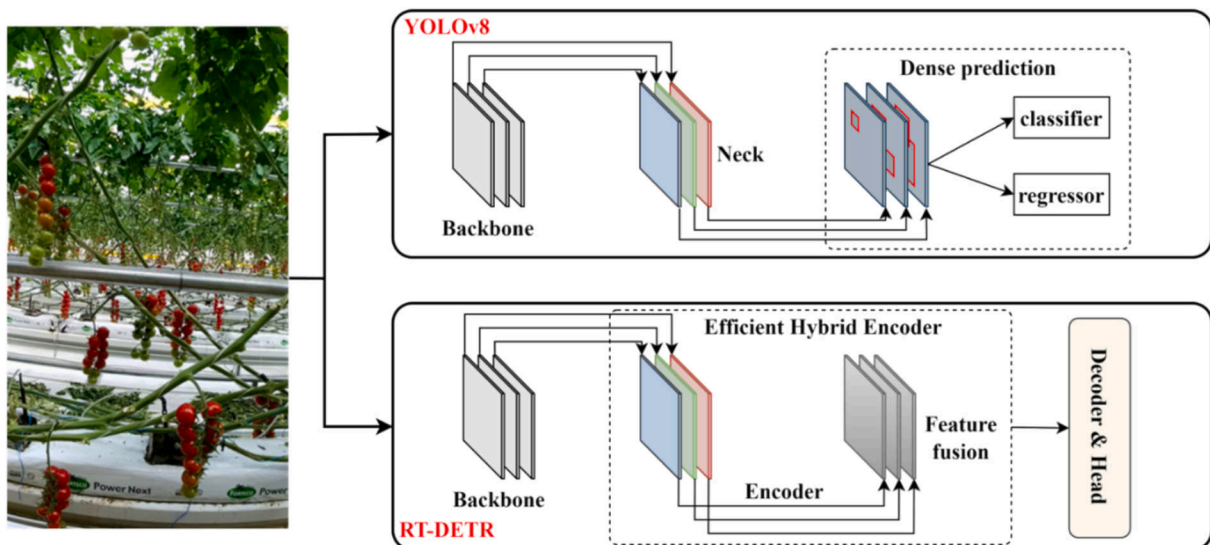


Fig. 3. Illustration of the tomato clusters detection procedure in the image sequence (i.e., video) based on the YOLOv8 and RT-DETR architectures.



Fig. 4. Main components of the multi-target tracking algorithm based on the tracking-by-detection scheme.

mainstream trackers. In the following, we first describe the four trackers in detail and then compare their capabilities.

2.3.1. Tracking via SORT

Simple Online Real-time Tracking (SORT) represents a real-time MOT algorithm [3], which is mainly constructed based on the algorithm uses Kalman filtering to predict the state of the detected bounding boxes. The state prediction results are intersected with the newly detected results by intersection-over-union (IoU) prediction and the similarity calculation and the data association problem is resolved via the Hungarian algorithm. Additionally, if the IoU between the predicted box and the newly detected bounding boxes is too small, the minimum IoU is applied to reject the allocation. The SORT algorithm only takes into account the movement of the bounding boxes without dealing with the ID switch due to the change in appearance. The state prediction for each bounding can be expressed as follows:

$$\mathbf{x}_t = [u_t, v_t, s_t, r_t, V_{u_t}, V_{v_t}, V_{s_t}] \quad (1)$$

where u_t and v_t represent the coordinates of the center point of the bounding box at moment t ; s_t and r_t in order denote the area and the aspect ratio, V_{u_t} and V_{v_t} represent the velocities of the center point, and V_{s_t} signifies the surface velocity.

2.3.2. Tracking via DeepSort

DeepSort [30] emerged to address issues pertinent to occlusion that could not be considered by the SORT algorithm chiefly achieves trajectory assignment only through IoU matching. In the face of occlusion, this could lead to frequent ID switches. Therefore, the DeepSort algorithm performs deep association metrics on objects by incorporating an appearance feature extraction module on the basis of CNN networks. Additionally, it optimizes the matching strategy by considering both the motion information and the appearance features. This DeepSort strategy could deeply influence subsequent trackers. In this algorithm, the predicted motion states are optimized:

$$\mathbf{x}_t = [u_t, v_t, \gamma_t, h_t, V_{u_t}, V_{v_t}, V_{\gamma_t}, V_{h_t}] \quad (2)$$

where u_t and v_t represent the coordinates the center of the bounding box at time t ; γ_t is the aspect ratio; h_t denotes the height; and V_{u_t} , V_{v_t} , V_{γ_t} , and V_{h_t} signify the velocities of the remaining four states.

2.3.3. Tracking via ByteTrack

ByteTrack, introduced by Zhang, Sun, et al. [35], presents a distinct yet efficacious approach to data association, discriminating that from conventional methods like DeepSort. The tracker initiates the association process by correlating high-confidence detection boxes with existing tracks. However, some tracks often fail to find matches, predominantly due to occlusions, motion blur, or significant size variations. To address these issues, the methodology subsequently associates lower-confidence detections with these unpaired tracks, thereby facilitating object retrieval in frames with lower scores. The underlying logic assumes that low-scoring bounding boxes frequently encapsulate instances marred by substantial occlusions or motion blur, rendering appearance features unreliable. As a result, the algorithm adopts a dual-stage matching strategy, leveraging high-confidence detections for identification while avoiding reliance on potentially misleading low-confidence features. This selective approach not only enhances the robustness of target identification but also contributes to a reduction in computational overhead, thereby optimizing efficiency.

2.3.4. Tracking via BotSort

BotSort [38] is also a SORT-like multi-object-tracking algorithms, with key innovations that include the integration of the state-of-the-art FastID as the appearance feature extractor. Additionally, it features a camera motion compensation module to enhance tracking accuracy. This module reduces the Kalman filter state vector deviations arising from positional changes caused by the camera jitter. Similar to ByteTrack, it uses a two-step matching process for both high and low confidence detections. However, during the initial matching step, it integrates both IoU and Re-Identification (ReID) metrics to enhance data association robustness. Furthermore, in the BotSort state estimation matrix, the aspect ratio adjustment is reformulated to width. The specific formula of this algorithm reads:

$$\mathbf{x}_t = [u_t, v_t, w_t, h_t, V_{u_t}, V_{v_t}, V_{w_t}, V_{h_t}] \quad (3)$$

where: the width is w_t . The velocities of the width is V_{w_t} .

2.4. Counting method

Despite the continuous improvements in the performance of trackers, they still fail to meet the requirements of accurate tomato cluster counting. The main challenge lies in the frequent identity switches, leading to discontinuous tracked identifications of bunches. Utilizing the maximum ID value as the count would remarkably overestimates the actual number of bunches. As a result, we have adopted a region-based counting approach [23] to gauge the number of tomato clusters. This technique involves delineating a consistent region on each image frame. Each video frame displayed in Fig. 8 demonstrates a fixed mask region, which is designated as the fixed area. As the frames change, tomato bunches continuously enter this designated area. The coordinates of the upper left corner of the fruit cluster bounding boxes also enter this area accordingly. We determine whether each fruit cluster tracked in the current frame should be counted by checking whether the upper left coordinates of the fruit cluster are in the fixed area. Real-time counting is achieved by counting the number of tomato bunches IDs that have entered this fixed area. If a new cluster of tomato ID enters the area, one cluster should be added to the total number. In this way, the number of tomato cluster IDs that enter the region can be considered as the estimated number of fruit clusters. This method is capable of effectively enhancing the counting accuracy.

2.5. Evaluation metrics

To assess and compare the detection performance of different detectors, we utilize the evaluation metrics that are commonly used for object detection models. For this purpose, the average precision (AP) is employed to metric the performance of the detectors. Based on the precision (P) and recall (R) of the object detection model, AP is computed as follows:

$$P = \frac{TP}{TP + FP} \quad (4)$$

$$R = \frac{TP}{TP + FN} \quad (5)$$

$$AP = \int_0^1 P(R) dR \quad (6)$$

where: TP , FP and FN represent true positives, false positives, and false

negatives, respectively.

Traditionally, multiple object tracking accuracy (MOTA) serves as a fundamental metric in evaluating the performance of MOT-based systems. In line with the common practices, our study includes it as a critical metric for evaluating the tracker performance. In addition, to accurately measure the object counting accuracy among individual videos consisting k frames, we use the relative precision (P_{rel}) as a complementary metric. The mathematical formulas defining these two evaluative indices are described in the following form:

$$MOTA = 1 - \frac{\sum_k (m_k + FP_k + IDSW_k)}{\sum_k GT_k} \quad (7)$$

$$P_{rel[i]} = \left(1 - \frac{|N_{actual[i]} - N_{est[i]}|}{N_{actual[i]}} \right) \times 100\% \quad (8)$$

Where m_k , FP , and $IDSW_k$ denote the number of misses, false alarms, and identity switches at k frame, respectively, while GT_k represents the total number of ground truth objects at that time instant. i denotes the serial number of a test video, and N_{actual} and N_{est} indicate the actual and counted number of tomato clusters in the video, respectively.

P_{rel} attained for each video is compiled into a tabular format; nevertheless, variations in accuracy may appear as peaks or troughs, depending on factors such as video quality and the number of frames. To ensure fairness and consistency in assessing, we lessen these discrepancies by averaging the relative errors across multiple videos, resulting in an average relative error. Such calculations are mathematically provided as:

$$\overline{P_{rel}} = \frac{1}{N} \sum_{j=1}^N P_{rel[j]} \quad (9)$$

where, N denotes the total number of test videos, and j denotes the video serial number.

3. Result and discussion

This section first presents a comparison of the performance of the two object detectors. Then, the performance of the trackers is compared on fifteen test videos. Finally, a discussion is given on the application of the tracking counting strategy in tomato counting.

3.1. Tomato detection results

We utilized the metrics as defined in Eq. (6) to compare the performance difference between YOLOv8 and RT-DETR. With an IoU threshold of 0.5, both models achieve comparable performance see Table 2). However, RT-DETR outperforms YOLOv8 under more rigorous conditions. Fig. 5 illustrates the detection outcomes of each model when processing two frames from a test video. Specifically, Fig. 5a,b demonstrate that YOLOv8 misclassifies a single cluster of tomatoes as two objects under poor lighting, whereas RT-DETR correctly identifies it as a single cluster. Fig. 5c,d present the response of the models to background rows, where YOLOv8 erroneously incorporates background tomatoes in the current row count. Whereas RT-DETR effectively reduces this background interference and confidently identifies the desired clusters. In the subsequent analyses the tracking sensitivity analyses, RT-DETR was exclusively selected as the detector to minimize inaccuracies in detection affecting tracking results. To objectively assess the impact

Table 2

Comparisons of the detection results of the two object detection models.

Model	AP@0.5	AP@0.5:0.95	AP@0.75
YOLOv8	98.5 %	80.4 %	93.6 %
RT-DETR	98.5 %	81.2 %	94.9 %

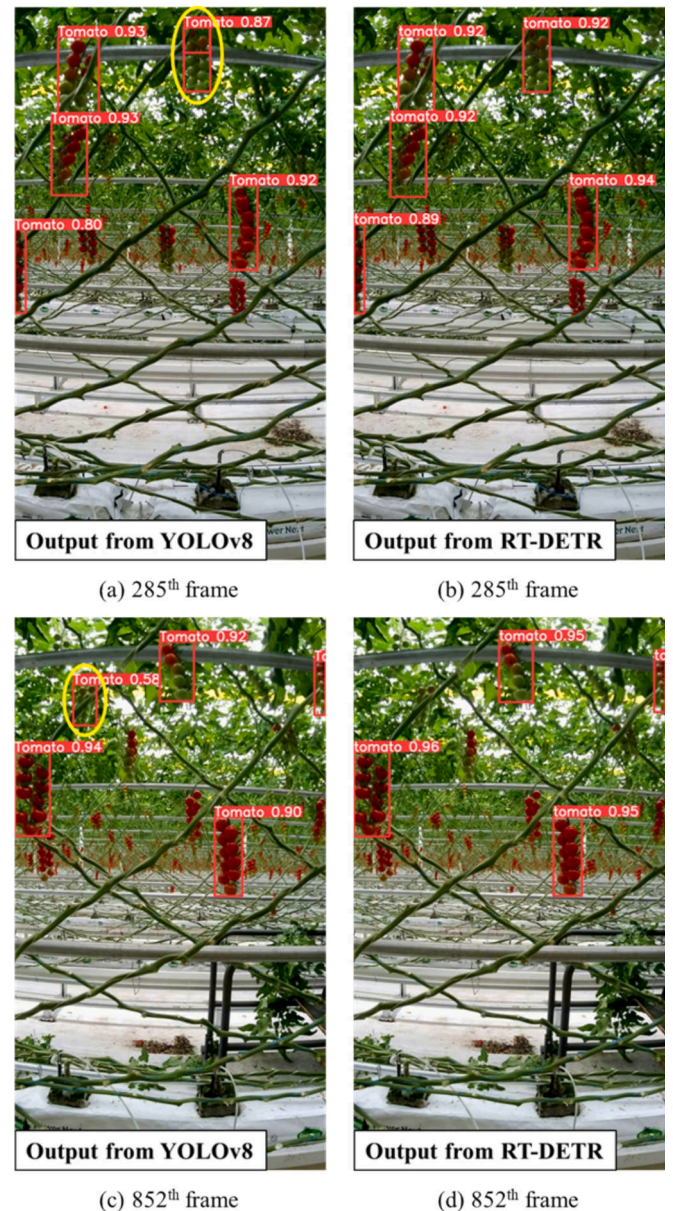


Fig. 5. Comparison of the outputs of YOLOv8 and RT-DETR models: (a) YOLOv8 recognizes a tomato bunch in the color change stage as two bunches, (b) The correct detection results from RT-DETR. (c) YOLOv8 misidentifies a bunch in the background, (d) The correct detection results from RT-DETR. (note: In all above subfigures, two examples of the 3rd video are provided).

on counting, counts obtained using YOLOv8 as the detector were also included for a thorough comparison, as detailed in Table 2 and Fig. 7.

3.2. Sensitivity analysis

Fig. 6a shows the result of the evolution of the number of fruit clusters estimated by the tracking algorithm compared to the actual number as a function of the number of frames. It is noteworthy that both the actual and estimated number increase steadily with the increasing number of frames, despite slight discrepancies between the two. Given the variation in frame counts across different video segments, this analysis recognizes potential impacts on the counting accuracy. To address issue, a correlation analysis of 15 test videos was utilized. As depicted in Fig. 6b, the results demonstrate a high determination coefficient ($R^2 = 0.95$) from the BotSort tracker, attesting to the tracker's proficiency in approximating the true fruit count. However, fluctuations

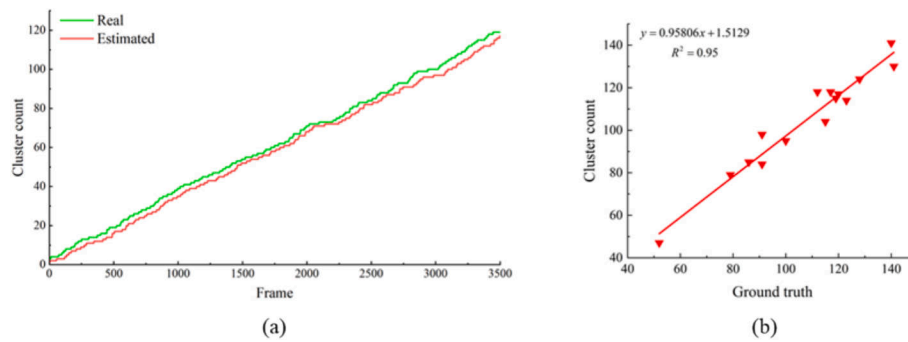


Fig. 6. Illustration of the cluster counting performance in a generic tracker: (a) Ground truth and tracking method estimation (note: the statistical results of the estimated and real value of algorithm using VT4 as a video example); (b) A linear regression between the number of clusters counted by tracker algorithm and that of the manual vision (note: the value in the Y direction of each dot in the graph represents the number of valid IDs counted by algorithms in a video segment, and the value in the X direction represents the number of tomato clusters that are manually counted).

in the frame rate in real-world applications could still affect the counting results. To evaluate the trackers' response to frame rate variations, a series of experiments were conducted. Deliberately reducing the frame rate to one-third, one-half, or maintaining the original rate (1:1) resulted in a corresponding reduction in the number of processed frames. Such changes could cause substantial positional shifts for a fruit bunch in sequential frames, thus making the tracking program more difficult to process. The case study revolves around VT4 video, with extensive benchmarks collected under various frame rates. Fig. 7 provides a detailed analysis of the effect of sampling frequency on the MOTA for understudied tracking algorithms. The plotted results reveal a consistent trend where performance reduces as the frame rates decreases. Notably, algorithms that incorporate Re-Identification (ReID) modules, such as DeepSort and BotSort, show a more severe performance degradation compared to the simpler SORT and ByteTrack algorithms that exhibit greater robustness in the presence of varying frame rates. Table A.1 presents a detailed statistical summary that includes ground truth (GT) counts, maximum ID (MaxID) obtained by the tracker, estimated counts based on region counting, and performance metrics such as IDF1 and MOTA for the understudied trackers. The optimal performance is presented to approach unity with higher values of IDF1 and MOTA. Notably, at a reduced frame rate of 10 fps, remarkable variances in MaxID and estimated counts are observed for DeepSort and BotSort, highlighting the increased sensitivity of CNN-based appearance feature extraction to changes in frame rate. The visualization in Fig. 8 reinforces

these findings, sorting frames by equivalent frame rate horizontally and tracker types vertically. By plotting the centers of tracked objects in multiple frames in a single snapshot, the continuity and trajectory consistency of the bounding box paths are visually assessed. At 30 fps, all trackers track every target well with coherent trajectories. When the frame rate is reduced to 15 fps, DeepSort misses a target, consistent with previous observations. At 10 fps, DeepSort's performance worsens and BotSort also loses a target. The SORT-based algorithm continues to track all targets but suffers significantly from persistence, whereas ByteTrack outperforms with its consistent tracking capability. These visual analyses, taken together, confirm the quantitative assessments from Fig. 7, and highlight the diverse resilience of trackers to frame rate changes.

3.3. Counting results

The performance of the counting system was mainly affected by both the detector and tracker components. Compared to YOLOv8, RT-DETR exhibited fewer false detections, which was consistent with a negative impact on the tracking efficacy (see Fig. 5). Despite this, several appropriate experiments were conducted via YOLOv8 and RT-DETR models at a frame rate of 30 fps as detectors, each paired with four distinct trackers. The MOTA and P_{rel} metrics for these pairings are provided in Table 3. The superior MOTA and P_{rel} were consistently achieved based on the RT-DETR model, except for DeepSort, which demonstrated an increasing trend for P_{rel} , although most likely due to an increased frequency of false detections by YOLOv8. This led to the production of more IDs and, consequently, a smaller final counting discrepancy when compared to DeepSort using RT-DETR. The trackers that have a ReID module, such as DeepSort and BotSort, ensure a higher MOTA score, indicating the potential of the ReID module to reduce ID switches. However, batches with analogous postures and ripening stages may confuse the appearance-based identification module and render it ineffective in certain scenarios. Conversely, trackers that rely solely on bounding box detection and motion prediction, which include SORT and ByteTrack (see Table A.1), can effectively distinguish bunches of various fruits regardless of appearance similarities due to the inability to overlap the locations of similar-looking tomato bunches in a single frame. Although BotSort, has a ReID module like DeepSort, the addition of a motion compensation feature in BotSort contributes to its superior identification accuracy and yields better counting results. The comparative bar charts of the counting results based on various detector-tracker combinations are plotted in Fig. 9, which demonstrate that each pair with RT-DETR outperforms those utilizing YOLOv8. Notably, ByteTrack and BotSort demonstrated lower error rates. Given the previous sensitivity analyses, we can rationally conclude that the combined utilization of RT-DETR and ByteTrack leads to the most accurate and robust counting algorithm.

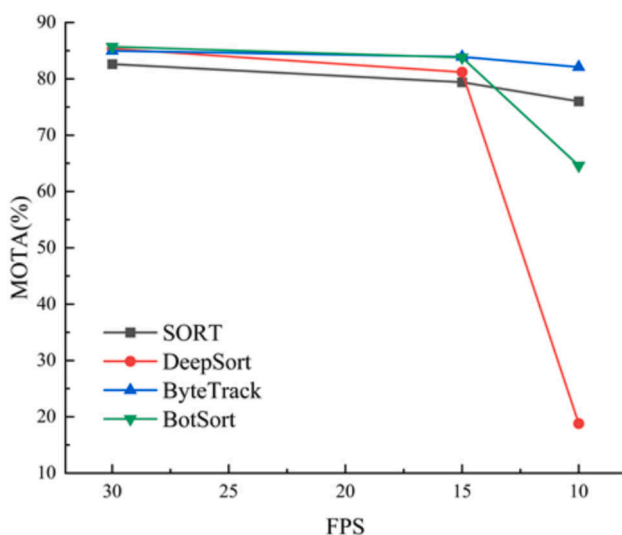


Fig. 7. Sensitivity analysis of various object trackers based on the MOTA-FPS graphs.

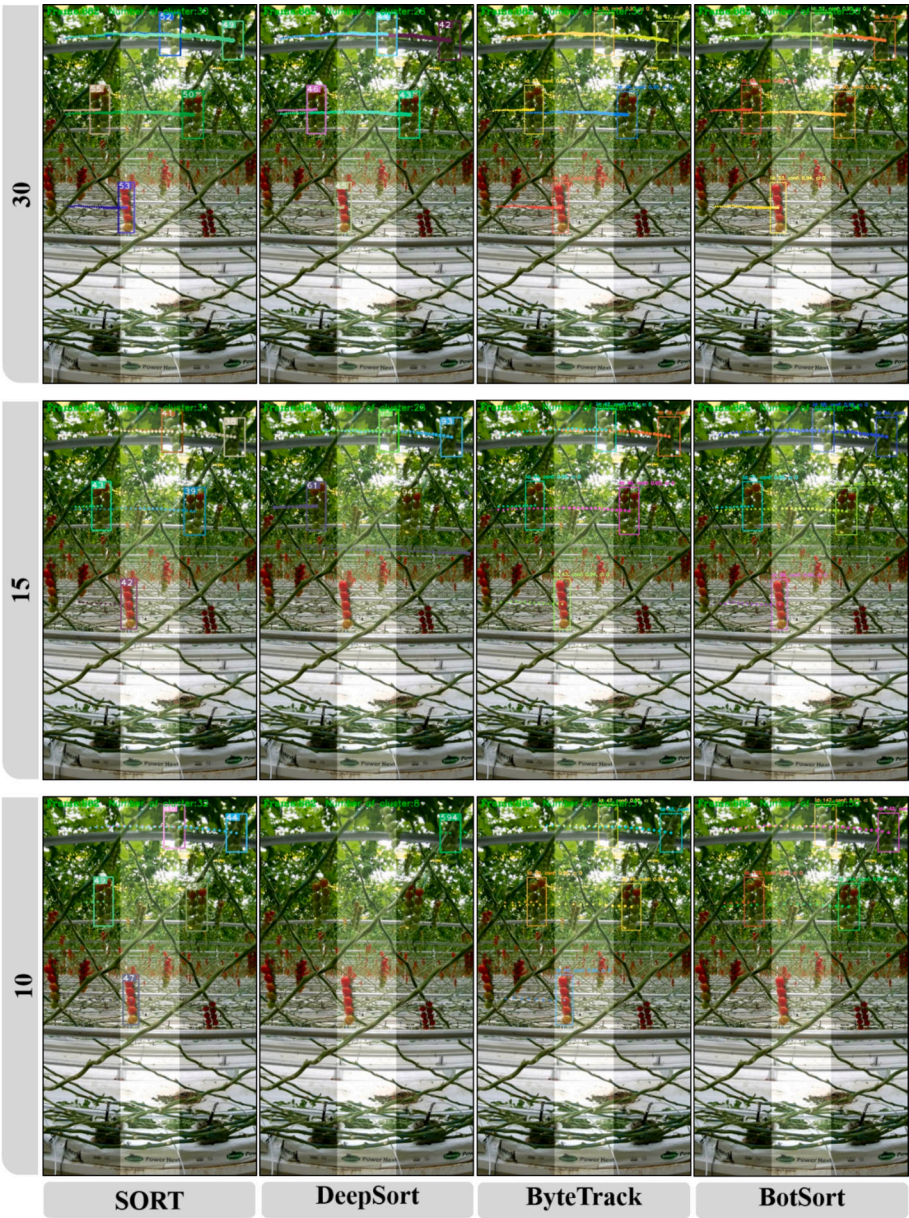


Fig. 8. Visualization comparison of various trackers (note: The illustrated results are from the output visualization of video segment VT4. The rows represent visualizations of the same frame rate at various trackers, whereas the columns denote visualizations of the same tracker at different frame rates. A fixed area of white mask for tomato bunch counting has been added to each image in the figure).

Table 3
Comparison of the predicted results after different combinations of detectors and trackers (note: Only the average MOTA and $\bar{P}_{rel}$ based on Eqs. (7)-(9) are provided).

Model	SORT MOTA	$\bar{P}_{rel}$	DeepSort MOTA	$\bar{P}_{rel}$	ByteTrack MOTA	$\bar{P}_{rel}$	BotSort MOTA	$\bar{P}_{rel}$
YOLOv8	80.1 %	94.0 %	82.1 %	91.1 %	81.4 %	94.4 %	82.0 %	93.5 %
RT-DETR	82.3 %	94.9 %	84.3 %	87.6 %	83.5 %	95.5 %	84.6 %	95.2 %

3.4. Discussion

Unlike traditional agricultural settings, modern greenhouse farming standardizes various agronomic elements, such as uniform growing platforms and fixed vehicular pathways (see Fig. 2b). This reduces the complexity of deploying agricultural robots in the diverse and often less structured agricultural fields. For the specific task of counting tomatoes in the growing area, it is essential that the device utilized could capture a complete tomato object. Hence, an OAK camera with an extended FOV

was implemented to collect visual data. The fixed position of the camera helps in the detector task to enhance accuracy in counting. During the tomato detection process, bunch detection commonly occurs in cases of leaf culling, where complete occlusions are rare and most instances involve partial ones. The advanced detector is highly skilled at identifying these clusters. The main factors that interfere with the detector include variations in lighting and the presence of tomatoes in the background ridge. These situations could lead to false detection by the detector and have an adverse impact on the tracker. Compared with

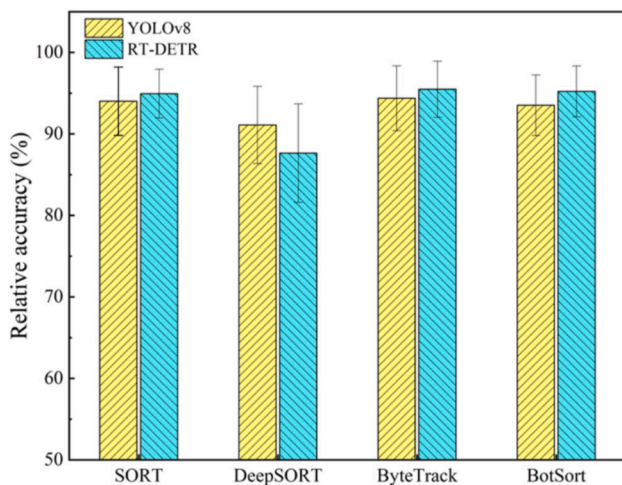


Fig. 9. Comparison of the tracking strategies (note: The plotted results represent relative errors of various tracking approaches based on the YOLOv8 and RT-DETR object detectors. Additionally, the bar plot represents the mean value, and the vertical line above the top of the bar signifies the standard deviation).

YOLOv8, RT-DETR possesses the advantage of reducing false detection. Although RT-DETR-based model has made significant optimization between speed and accuracy compared with the original DETR, it still exhibits a disadvantage in speed when compared with YOLOv8; however it is sufficient for real-time fruit cluster detection.

When tracking tomato clusters, different trackers exhibit dissimilar performances. In addition to the detector, the tracker also possesses three key components (feature extraction, similarity calculation, and data association). In general, more complex trackers, apart from the above key components, may also add other useful modules for target tracking, such as a camera motion compensation module, which may achieve better performance. When tracking general targets, factors such as occlusion, appearance similarity, and camera movement should be considered. In the scenario involved in the current investigation, a fixed camera could reduce the misinterpretation by the detector and basically keep the camera stationary, except for jolts when the camera moving. The standardized track design could enable the platform vehicle to achieve more stable driving speeds. These factors help to improve the tracking effects. However, the movement speed of the platform carrying the camera remains a certain influencing factor that cannot be ignored. In practical applications, the platform may experience deceleration due to different situations, which could lead to inconsistent image quality. These conditions not only affect the performance of the detector but also the performance of the tracking algorithms. For detectors, differences in image quality directly lead to unstable detection results, such as false positives and false negatives. Object detectors could mitigate the negative effects of speed variations to some extent through various data augmentation techniques. As for trackers, the algorithms we compared mainly rely on the Kalman filtering to predict the position of the bounding boxes. Therefore, excessively fast or slow speeds will cause fluctuations in the prediction of bounding box positions, which, in turn, affects the data association phase. The results listed in Table A.1 reflect the tracking results in the presence of various frame rates and indirectly exhibit the impact of speed variations on the tracking performance, with noticeable discrepancies in robustness observed among various trackers. Overly complex component designs, although increasing the value of the MOTA metric, have less impact on the counting results. The frame rate fluctuations leading to worse tracking outcomes are similar to the observations of Villacrés et al. [28]. However, contrary to their opinion, the decline is more pronounced in ReID-based trackers. The benefits of the ReID and camera movement compensation modules are limited. Moreover, in the case of frame rate fluctuations, the ReID module actually incorporates an increase of more ID conversions, which is not

beneficial for effective deployment in real-world scenarios. This may be due to the platform moving along a straight path, where overly complex gain modules cannot adapt to simple linear movements, and overly similar appearances of the fruit bunches might disturb data association through the ReID module, resulting in more ID switches. Although BotSort achieved the best MOTA metric, its robustness is weaker than SORT and ByteTrack. Considering the speed of the algorithm, SORT and ByteTrack also better meet the needs of real-time performance requirements.

As a result, the ByteTrack tracker, integrated with the RT-DETR detector and without the ReID module, achieves the best counting effect and the most robust tracking results. In future research, we plan to further integrate the previously maturity estimation methods for fruit clusters into the counting algorithm. Our goal is to deploy the algorithms on a standalone mobile inspection platform and transmit the data to the control center via the network, realizing real-time counting and maturity estimation, which will help to quantitative produce of tomatoes.

4. Conclusions

To conduct a comprehensive assessment of the performance and application prospects of based on the tracking by detection approaches applied to the yield estimation for greenhouse tomato clusters, this study evaluated a various of methods based on video data. To evaluate the performance, we constructed a public large-scale MOT dataset on tomato bunches. To this end, we evaluated several metrics for both detectors and trackers, specifically comparing two real-time detectors—YOLOv8 and RT-DETR—and four trackers—SORT, DeepSort, ByteTrack, and BotSort. Under various video frame sampling frequencies, SORT and ByteTrack demonstrate better robustness. While ReID-based trackers such as DeepSort and BotSort exhibit superior performance under constant sampling. Using RT-DETR as the detector, BotSort produces the highest MOTA of 84.6 %, and ByteTrack ensures the highest counting accuracy at 95.5 %. In addition, sensibility analysis to frame rate fluctuations reveals that based on non-ReID trackers (SORT and ByteTrack) outperform their counterparts in tracking stability. In near future, the proposed algorithms have strong potentials to be deployed for in autonomous inspection platforms to facilitate real-time counting and aid in managerial decisions for tomato production.

5. Ethics Statement

Not applicable: This manuscript does not include human or animal research. If this manuscript involves research on animals or humans, it is imperative to disclose all approval details.

CRedit authorship contribution statement

Zhongxian Qi: Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Tianxue Zhang:** Funding acquisition, Writing – review & editing. **Ting Yuan:** Writing – review & editing, Supervision. **Wei Zhou:** Validation, Investigation. **Wenqiang Zhang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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that inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, or service.

Appendix

See Table A.1.

Table A.1

This comparative analysis evaluates the impact of the changed frame sampling rate on the tracker metrics, via VT4 video as a case study. The original frame rate is 30fps, sampled at 1/1, 1/2 and 1/3 of the initial sampling rate (i.e., 30fps, 15fps and 10fps respectively). Higher values of the MOTA and IDF1 indicate better tracker performance.

FPS	Tracker	GT	MaxID	Count	IDF1↑	MOTA↑
30	SORT	120	197	117	89.7 %	82.6 %
	DeepSort	120	163	107	86.2 %	85.4 %
	ByteTrack	120	187	117	91.1 %	85.0 %
	BotSort	120	198	117	91.4 %	85.7 %
15	SORT	120	169	112	88.3 %	79.4 %
	DeepSort	120	183	114	88.8 %	81.2 %
	ByteTrack	120	169	114	91.3 %	83.9 %
	BotSort	120	269	117	90.5 %	83.8 %
10	SORT	120	168	113	86.3 %	76.0 %
	DeepSort	120	2479	28	30.4 %	18.8 %
	ByteTrack	120	178	112	90.2 %	82.1 %
	BotSort	120	607	195	74.4 %	64.6 %

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Ting Yuan is a vice professor at China Agriculture University. He received his Bachelor's degree in College of Information and Electrical Engineering from China Agricultural University in 2004, followed by a doctor's degree in College of Engineering from China Agricultural University in 2012. His research interests include machine vision and agricultural robots. Ting Yuan can be contacted via his institutional email: yuan-ting122@cau.edu.cn.



Wei Zhou is a student at China Agriculture University. He received his Bachelor's degree in Mechanical Engineering from Northeast University in 2021, followed by a Master's degree in College of Engineering from China Agriculture University in 2023. His research interests include microfluidics and agricultural robots. Wei Zhou can be contacted via his institutional email: zhouwei@cau.edu.cn.



Zhongxian Qi is a student at China Agriculture University. He received his Bachelor's degree in College of Engineering from China Agricultural University in 2016, followed by a Master's degree in College of Engineering from China Agriculture University in 2019. His research interests include machine vision and agricultural robots. Zhongxian Qi can be contacted via his institutional email: qzxcau@cau.edu.cn.



Wenqiang Zhang is a vice professor at China Agriculture University. He received his Bachelor's degree in College of Engineering from China Agricultural University in 2007, followed by a doctor's degree in School of Mechanical Engineering & Automation from Beihang University in 2012. His research interests include microfluidics and agricultural robots. Wenqiang Zhang can be contacted via his institutional email: zhangwq@cau.edu.cn.



Tianxue Zhang (Member, IEEE) Tianxue Zhang received her B. Eng. degree in electrical engineering from Beijing Institute of Technology, Beijing, China, in 2014, and her Ph.D. degree in mechanic and automation engineering from the Chinese University of Hong Kong, Hong Kong, China, in 2020. She was a Postdoctoral Research Fellow with the Institute of Medical Robotics, Shanghai Jiao Tong University, Shanghai from 2020 to 2022. She is currently an Assistant Professor with the School of Mechanical Engineering and Automation of Beihang University, Beijing, China. Her research interests include agricultural robots, and medical robots.